

COVER LETTER

August 14, 2026

Adrian Baranchuk, MD

Editor-in-Chief

Journal of Electrocardiology

Dear Dr. Baranchuk,

Please consider our manuscript, "**Record-Level Evaluation of a Morlet CNN-Conformer for Five-Class ECG Beat Classification**," for publication as a **Research Paper** in the *Journal of Electrocardiology*.

This study evaluates a Morlet wavelet CNN-Conformer for five-class ECG beat classification using a record-level holdout of seven MIT-BIH Arrhythmia Database recordings. We compare the proposed model with CNN-attention, CNN-LSTM, and CNN-only architectures under the same split and report aggregate, class-specific, and error-level results.

The CNN-Conformer produced the highest aggregate performance among the four evaluated models (accuracy 0.60, macro-F1 0.26, and weighted-F1 0.68). Importantly, our analysis shows that the gain was driven mainly by Normal and ventricular ectopic beats, while supraventricular ectopic and Fusion recognition remained near zero. This record-level, failure-aware evaluation is intended to provide a cautious and reproducible account of what the current model supports before broader clinical claims are made.

The manuscript is well aligned with the journal's focus on computerized electrocardiology, arrhythmia analysis, and clinically meaningful interpretation of ECG algorithms. The MIT-BIH data are publicly available, and the source code, experiment configurations, trained models, and prediction arrays are provided in a fixed public GitHub release linked in the manuscript.

This manuscript is original, has not been published, and is not under consideration by another journal. The authors declare no competing interests and no specific external funding. The study used only publicly available, fully de-identified data; no new human or animal data were collected. A generative-AI assistance disclosure is included directly before the references.

Thank you for your consideration.

Sincerely,

Alireza Abbaszadeh

Corresponding author

Department of Computer Engineering, Islamic Azad University

Mashhad, Iran

Email: alireza.abbaszadeh8558@iau.ir

Telephone: +98 915 206 2041

On behalf of Alireza Abbaszadeh and Vahid Torkzadeh